A Unified Multitemporal Learning Framework for Semantic Change Detection in Remote Sensing Images

Yuhang Zhang, Yang Zhao, Wuxia Zhang, *Associate Member, IEEE*, and Lei Chen

***Abstract*—Semantic change detection (SCD) in remote sensing time series is a crucial component of land cover/land use (LCLU) dynamic monitoring. It enables the accurate identification of change areas and holds significant application value. However, existing remote sensing time-series SCD tasks are relatively limited in scope, failing to fully capture the synergistic relationship between semantic changes and temporal evolution. To alleviate the above issues, this article proposes a unified multitemporal learning framework that integrates semantic segmentation (SS) with multitemporal binary change detection (MBCD). This approach enables the collaborative modeling of multitemporal change patterns and semantic information. This framework primarily consists of three components: the SS encoder (SSE), the MBCD Block, and the multitemporal learning decoder (MTLD). By introducing the MBCD Block, it effectively integrates multitemporal semantic features to capture dynamic change information. The wavelet spatial-frequency block (WSFB) and joint multiloss supervision proposed in the MTLD enhance the model's fine-grained semantic perception capabilities and few-shot learning capability. Furthermore, we resampled and annotated the Hi-UCD mini and WUSU datasets to construct two multitemporal SCD datasets, providing data support for model training and evaluation. The experimental results show that the proposed method can achieve more accurate SCD in multitemporal remote sensing image (RSI). The source code is available at https://github.com/YuhangZh/UMTL-Net**

***Index Terms*—Remote sensing time series, semantic change detection (SCD), time series feature modeling.**

I. Introduction

CHANGE detection (CD) in remote sensing images (RSIs) refers to comparing RSI of the same area at different time periods to identify changes in land cover/land use (LCLU) over time [1]. These changes can be caused by natural factors or human activities. CD, as an important basic application in the field of remote sensing, is becoming an important tool for monitoring changes in the Earth's surface and the environment. This provides strong support for multiple fields, including agriculture [2], [3], [4], environmental monitoring [5], [6], urban development [7], [8], and disaster management [9].

CD in RSI is typically divided into binary CD (BCD) and semantic CD (SCD) [10]. BCD identifies whether LCLU has changed, while SCD focuses on the specific type of changes, providing "from-to" information for the changed area. Compared with BCD, SCD can achieve more accurate identification and classification of LCLU. Therefore, SCD has higher application value and research significance in remote sensing research [11], [12].

Currently, most research on SCD algorithms still focuses primarily on scenes of bi-temporal RSI, with only a few studies attempting to extend the methods to multitemporal RSI (i.e., more than two time points). With the continuous advancement of satellite, UAV, and sensor technologies, an increasing number of regions can now obtain RSI sequences covering multiple time points [13], [14], [15]. In this context, bi-temporal imagery has limitations in multitemporal remote sensing analysis. Because it can only capture changes between two points in time and cannot fully reflect the continuity and dynamic evolution of surface changes.

Fig. 1 illustrates the differences between bi-temporal and multitemporal RSI in SCD. The white regions denote unchanged areas, for which semantic categories are not considered. Bi-temporal SCD methods use only two observations, e.g., 2017 and 2019, to identify LCLU changes. While effective for abrupt changes, they may fail to capture gradual or staged transitions. For example, in the road reconstruction case shown in Fig. 1(a), the area marked by the red circle changes from "road" to "bare land" and then back to "road," indicating an intermediate renovation process. Similarly, in the construction case shown in Fig. 1(b), bi-temporal SCD captures only the pre- and post-construction states, overlooking intermediate construction stages. In contrast, multitemporal SCD can better model object evolution trajectories and identify detailed "from-to" semantic changes, making it more suitable for applications such as urban expansion [16], [17], crop growth monitoring [18], and long-term environmental monitoring [19], [20].

Some scholars have already used multitemporal RSI to study CD. Yuan et al. [21] proposed the large-scale temporal dataset FUSU and modeled temporal information by fusing intermediate temporal image features with bi-temporal RSI alignment. However, their method is still limited to bi-temporal SCD and fails to fully utilize the semantic information of the intermediate phase. Hafner et al. [22] proposed a

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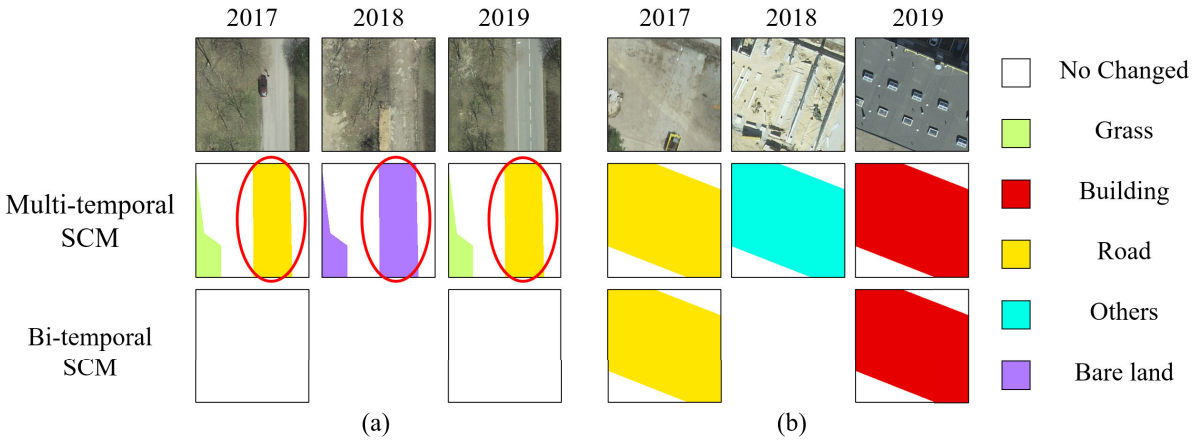


Fig. 1. Difference between bi-temporal and multitemporal remote sensing SCD. SCM stands for semantic change map. (a) Road reconstruction. (b) Construction site.

continuous urban CD method. This method improves the temporal continuity of BCD, but lacks semantic discrimination capabilities. Zheng et al. [23] proposed a method for generating multitemporal remote sensing change data based on generative modeling. Although this method can simulate change processes, the generated LCLU types are limited and cannot cover complex semantic spaces. Li and Wu [24] proposed a multitask network that can locate the occurrence time of change areas in time series RSI, but it cannot accurately capture the LCLU type in the change areas. Vincent et al. [25] designed a model architecture that utilizes long-term time series information. However, this method splits the SCD task into separate processes, failing to achieve deep integration between multitemporal change information and semantic information.

Currently, research on SCD in remote sensing time series is still relatively limited. A review of existing studies reveals that its development is primarily constrained by the following three key challenges that remain unresolved: first, there is a lack of a comprehensive information integration strategy. Currently, there is no unified multitemporal remote sensing SCD framework that can integrate temporal modeling and semantic detection. This limits the effective characterization of spatial–temporal change patterns and the accurate identification of semantic information in remote sensing time series. Second, multitemporal RSI (applicable to three or more temporal phases) has great potential to capture dynamic changes on the Earth's surface. However, most existing studies fail to fully exploit this potential. They often overlook the effective modeling of temporal information and lack in-depth exploration of temporal features. Furthermore, existing feature representations are largely confined to the spatial domain, failing to integrate complementary information from other domains (such as the frequency domain). This information is crucial for accurately capturing change boundaries and texture.

To address the above challenges, we propose a unified multitemporal learning framework for SCD in multitemporal RSI. By jointly learning semantic segmentation (SS) and multitemporal BCD (MBCD), the proposed framework enhances the model's ability to capture remote sensing spatial–temporal change patterns and recognize semantic information. By introducing the MBCD Block, we fuse multitemporal semantic features to effectively extract dynamic semantic changes in time series. Combining the discrete wavelet transform (DWT) and attention, we designed the wavelet spatial-frequency block (WSFB) to enhance the model's fine-grained perception of changing regions. Finally, we resampled and annotated the Hi-UCD mini [26] and WUSU [27] datasets, constructed two multitemporal SCD datasets, and verified the effectiveness of the method in experiments.

The main contributions are summarized as follows.

1) We propose a unified multitemporal learning framework that jointly models SS and MBCD, and improves the model's perception of temporal changes through multiloss collaborative optimization.
2) We propose the MBCD block, which utilizes Mamba to fuse multitemporal semantic features from multiple perspectives, enabling effective extraction of dynamic semantic changes in time series.
3) Within the multitemporal learning decoder (MTLD), we propose the WSFB and joint multiloss supervision to enhance semantic representation and improve the model's ability to capture semantic relationships between changed and unchanged regions, thereby improving few-shot learning performance.
4) We construct two multitemporal SCD datasets for model training and evaluation by resampling and annotating two remote sensing datasets (Hi-UCD mini and WUSU). The generalizability of the proposed method was then validated on the DynamicEarthNet dataset.

II. Related Work

A. Deep Learning-Based SCD Method

Currently, most deep learning-based SCD methods generally adopt an encoder–decoder architecture. Fig. 2 shows the current mainstream SCD framework. Although the three methods differ structurally, their core ideas are basically the same. Two encoders (Encoders 1 and 2) encode the input bi-temporal RSI and generate corresponding land cover maps (LCMs). Encoder 3 or Convs are used to generate a binary change map (BCM). At each pixel level, BCM predicts whether a change will occur, and the type of change is defined by the category predicted by LCM at that location.

Fig. 2(a) shows the framework of HRSCD-str.4 [11], where the semantic encoder features and the BCD decoder fea-

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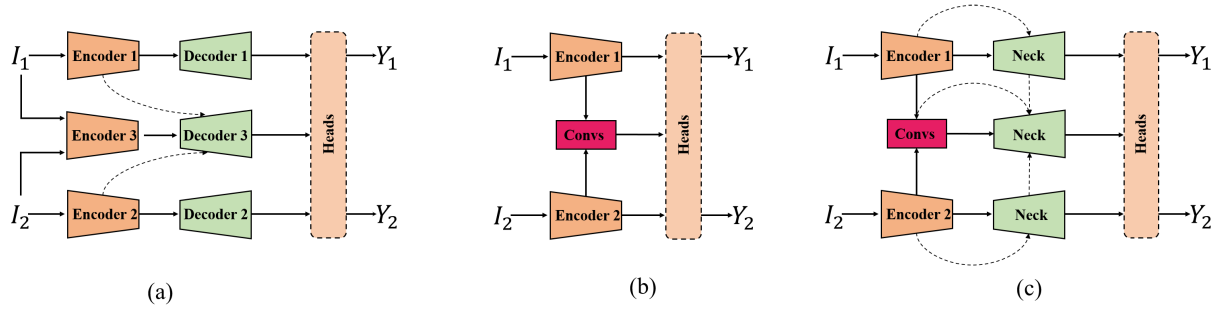


Fig. 2. Framework diagrams of different SCD methods. (a) HRSCD-str.4. (b) SSCD-l. (c) TED.

tures are independently encoded. In contrast, Fig. 2(b) shows SSCD-l [28], which discards Encoder 3 from HRSCD-str.4. and uses Convs to fuse the semantic features of bi-temporal images to generate BCDM. This design avoids redundant encoding of image semantic information, thereby simplifying the model structure.

Fig. 2(c) shows the framework of TED [29]. This method introduces a Neck block based on SSCD-l, further enhancing the spatial–temporal expression capabilities of features, thereby improving model performance. Currently, most SCD methods [30], [31], [32], [33], [34], [35], [36] are similar to this framework.

However, the above methods are still based on research using bi-temporal RSI. These methods are difficult to comprehensively capture the dynamic process of LCLU changes. In contrast, multitemporal SCD can more comprehensively characterize the dynamic process of LCLU evolution over time. It can also accurately pinpoint the timing of changes, providing temporal support for change analysis.

B. Mamba-Based CD Method

With the development of the state space model (SSM) from S4 to S6 [37], the Mamba model has been widely used in natural language processing tasks due to its linear complexity and comparable modeling ability to Transformer. In recent years, with the introduction of Mamba models in the field of vision, such as Vision Mamba (VMamba) [38], [39]. Mamba models have also been widely used in image processing tasks and have achieved good results.

Currently, the VMamba model has begun to be applied to RSI processing tasks. Chen et al. [40] proposed the ChangeMamba model, which achieves interaction between multiple spatiotemporal features through three spatiotemporal relationship modeling mechanisms, thereby obtaining accurate change information. The CDMamba model proposed by Zhang et al. [41] effectively integrates global and local features through ConvMamba, improving BCD performance. Liu et al. [42] proposed the CD-STMamba model, which interacts with image features from multiple dimensions to achieve CD. Zhao et al. [43] proposed the ST-Mamba model, which utilizes feature maps with unified background features to guide multiscale and cross-channel spatial–temporal feature fusion. Huang et al. [44] proposed Mamba semantic alignment networks, which effectively correct spatial misalignment issues in feature aggregation by integrating learnable semantic offset maps.

However, the above CD methods based on Mamba are mainly aimed at bi-temporal RSI. Due to the inherent lack of continuous time series features in bi-temporal images, it is difficult to capture long-term trends. Therefore, this article focuses on modeling temporal features in multitemporal RSI. The aim is to use visual state space (VSS) module to deeply explore and express temporal change patterns in RSI, thereby further improving the accuracy of SCD.

III. Method

A. Preliminary Work

SCD tasks can typically be decomposed into two subtasks: single-time-temporal SS and BCD. For single-image SS, bi-temporal SCD and multitemporal SCD share consistent task definitions. However, they differ in BCD tasks. For existing bi-temporal SCD methods, since only two images are available, BCM generation can directly reference bi-temporal BCD. Its modeling form can be expressed as

$$BCM = f(I_1, I_2) \tag{1}$$

where I_1 and I_2 represent bi-temporal RSI.

However, multiple methods exist for generating BCM from multiple images in multitemporal SCD tasks. In this section, we specifically analyzed two seemingly intuitive yet flawed multiphase scaling strategies.

1) Strategy 1 (Direct Comparison Method): As shown in Fig. 3(a), the first method is the direct comparison method. This strategy transforms multitemporal SCD into mutually independent SS tasks. First, use the encoder–decoder to process the multitemporal RSI $I_1 - I_3$ separately to generate the corresponding SS Maps (SSM_1, SSM_2, and SSM_3). Finally, SCD is achieved by identifying the changed areas through comparison of the three sets of SSM. Its modeling form can be expressed as

$$BCM = XOR(SSM_1, SSM_2, SSM_3, \ldots, SSM_n). \tag{2}$$

However, this strategy proves ineffective in practical applications, primarily due to the following issues: 1) it fails to explicitly model changes; 2) SCD results are entirely dependent on the segmentation accuracy of SSM; 3) errors are further amplified during the final comparison stage; and 4) it cannot characterize the continuous temporal evolution process.

2) Strategy 2 (Separate SS and BCD): As shown in Fig. 3(b), the second method is inspired by the bi-temporal SCD framework (Fig. 2). We also split the multitemporal SCD into two subtasks: BCD and SS. This framework decomposes multitemporal tasks into multiple adjacent bi-temporal problems, generating BCD maps: $BCM_1 = f(\boldsymbol{F_1}, \boldsymbol{F_2}), BCM_2 =$

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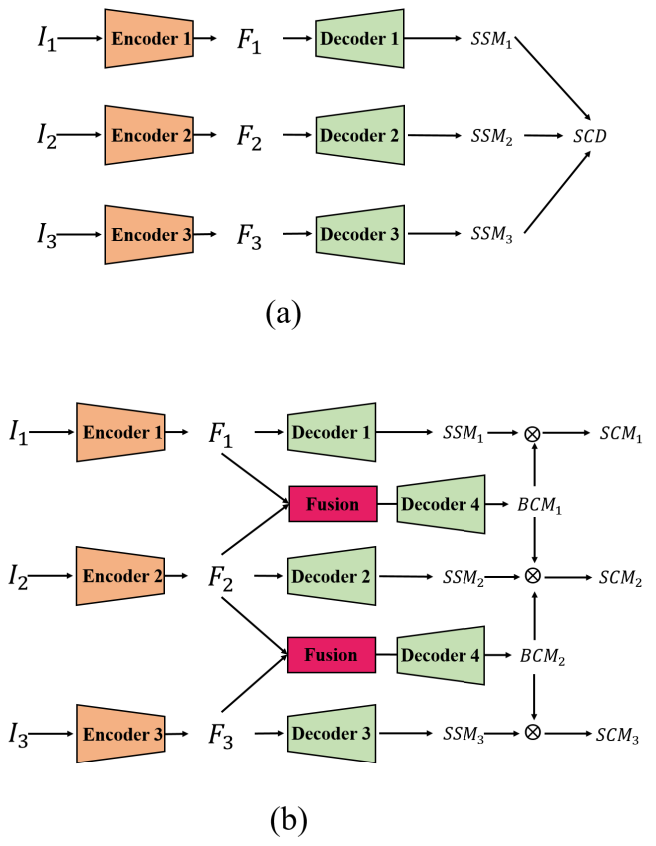


Fig. 3. Taking three temporal as an example, two different SCD strategies are illustrated. (a) Strategy 1: Direct comparison method. (b) Strategy 2: Separate SS and BCD. The abbreviations SSM, BCM, and SCM denote semantic segmentation maps, binary change maps, and semantic change maps, respectively.

$f(\boldsymbol{F_2}, \boldsymbol{F_3}), BCM_3 = f(\boldsymbol{F_3}, \boldsymbol{F_4}), \ldots, BCM_n = f(\boldsymbol{F_{n-1}}, \boldsymbol{F_n})$. Finally, multiply BCM by the predicted SSM to mask the unchanged areas and obtain the Semantic Change Map (SCM_1, SCM_2, SCM_3...).

However, our preliminary experiments revealed the following: 1) errors gradually accumulate over time; 2) global temporal dependencies are not explicitly modeled; and 3) as the number of phases increases, the model structure becomes increasingly complex.

3) Unified Multitemporal Learning Framework: As shown in Fig. 4, we propose a unified multitemporal learning framework, which primarily consists of the SS encoder (SSE), MTLD, and the MBCD block.

First, SSE encodes the multitemporal RSI I_1, I_2, and I_3, generating deep semantic features F_1, F_2, and F_3. The MBCD block then fuses these features to capture dynamic change information, generating a multi-BCM (MBCM). MTLD decodes the multitemporal image features to generate SSM_1, SSM_2, and SSM_3. Finally, multiplying MBCM by SSM_1, SSM_2, and SSM_3 yields the SCD results SCM_1, SCM_2, and SCM_3. These results not only mask unchanged areas in multitemporal RSI but also retain the semantic information of LCLU in changed areas.

Under this paradigm, we constructed a unified MBCM and integrated features from all temporal phases for joint modeling. At this point, the modeling object shifts from "whether changes occur at two points in time" to "the continuous evolutionary trajectory of feature semantics across the

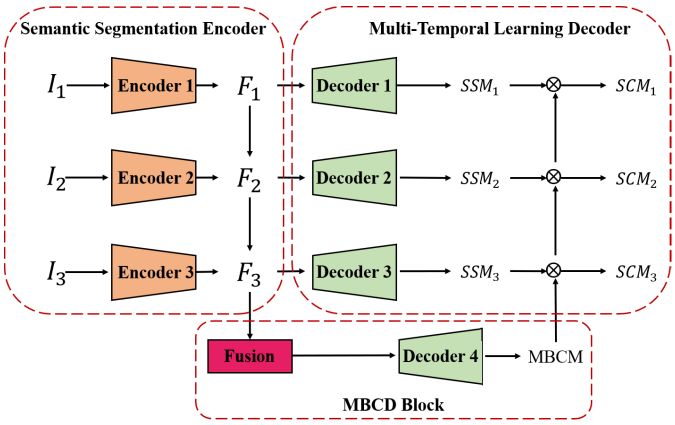


Fig. 4. Unified multitemporal learning framework. The abbreviations SSM, MBCM, and SCM denote semantic segmentation maps, multitemporal binary change maps, and semantic change maps, respectively.

temporal dimension." Its modeling form can be expressed as

$$MBCM = f(\boldsymbol{F_1}, \boldsymbol{F_2}, \boldsymbol{F_3}, \ldots, \boldsymbol{F_n}). \tag{3}$$

Specifically, the MBCM labels are obtained by performing an XOR operation on the semantic labels of multitemporal images.

The advantages of this approach are: 1) it can be naturally extended to longer time series; 2) it does not rely on the pairwise decomposition of adjacent time phases; and 3) it possesses continuous time series modeling capabilities.

B. SS Encoder

As shown in Fig. 5, this framework diagram illustrates the encoder architecture of the unified multitemporal learning framework. The purpose of this encoder is to efficiently extract semantic features from multitemporal RSI.

First, multitemporal RSI T_1, T_2, and T_3 are input into the encoder, and the spatial–temporal features of the images are gradually extracted through multiple VSS blocks. In this process, the resolution of image features gradually decreases, while the number of channels gradually increases, thereby generating feature representations $\boldsymbol{F_j}^i$ at different levels. This encoder framework can effectively capture detailed information in RSI and model long-distance spatial–temporal dependencies.

1) VSS Block: In the encoder, the VSS block is the key component, which mainly originates from SSM. SSM is a type of linear time-invariant (LTI) system that maps the input signal $x(t) \in \mathbb{R}$ to the output response $y(t) \in \mathbb{R}$ by introducing the hidden state $h(x) \in \mathbb{R}^N$. It is represented by the following linear ordinary differential equation (ODE):

$$\begin{aligned} h'(t) &= \boldsymbol{A}h(t) + \boldsymbol{B}x(t) \\ y(t) &= \boldsymbol{C}h(t). \end{aligned} \tag{4}$$

To integrate continuous-time SSM into deep models, it must first be discretized. By introducing a time scale parameter Δ and using the zero-order hold (ZOH) method to discretize the continuous parameters $\boldsymbol{A}$ and $\boldsymbol{B}$, we obtain $\bar{\boldsymbol{A}}$ and $\bar{\boldsymbol{B}}$

$$\begin{aligned} \bar{\boldsymbol{A}} &= exp(\Delta \boldsymbol{A}) \\ \bar{\boldsymbol{B}} &= (\Delta \boldsymbol{A})^{-1}(exp(\Delta \boldsymbol{A}) - \boldsymbol{I})(\Delta \boldsymbol{B}). \end{aligned} \tag{5}$$

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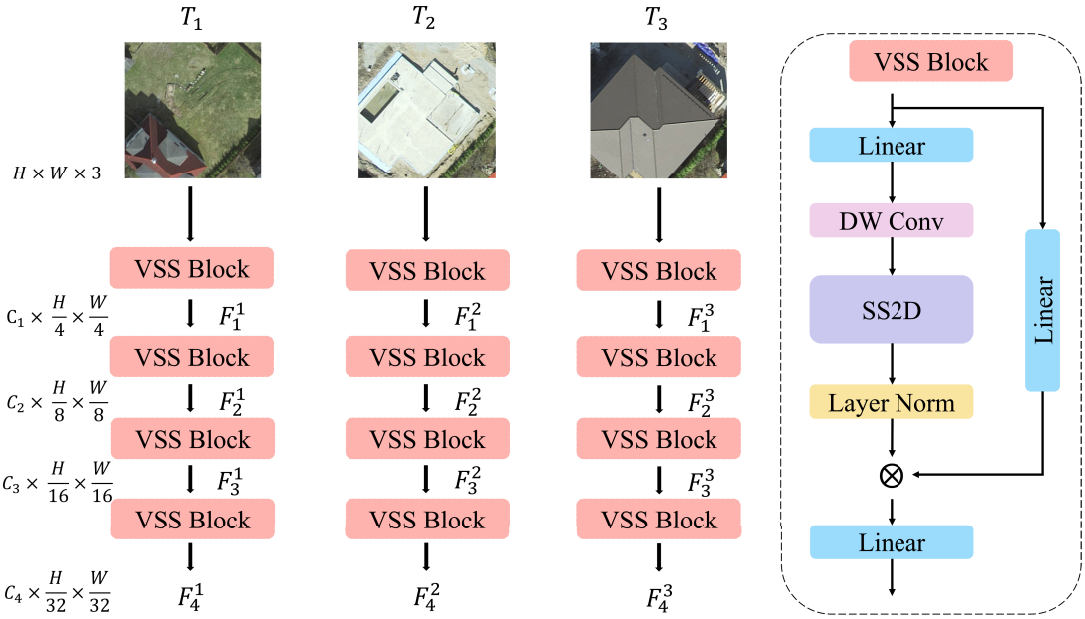


Fig. 5. SSE architecture of the unified multitemporal learning framework. The abbreviations VSS block, SS2D, and DW Conv denote visual state space block, 2-D-selective-scan, and depthwise convolution.

After discretization using (4) and (5) can be represented as

$$h_t = \bar{\boldsymbol{A}} h_{t-1} + \bar{\boldsymbol{B}} x_t$$
$$y_t = \boldsymbol{C} h_t. \tag{6}$$

However, the above parameters are fixed for different inputs. To address this issue, Mamba introduces a selective scanning mechanism that dynamically adjusts model parameters based on input, thereby selectively retaining or forgetting information.

In addition, a hardware-aware algorithm was developed for deep learning to optimize GPU resource utilization in recursive computation models. This enables Mamba to maintain linear-level computational overhead when processing long sequence data, thereby balancing efficiency and performance.

C. MBCD Block

Since a single RSI lacks explicit temporal information, it is difficult to directly reflect the dependencies between different time points. Therefore, in the feature extraction stage, by concatenating multitemporal image features, each set of features can implicitly contain temporal information. Since image features do not have strict temporal constraints, they can be concatenated from multiple dimensions. This approach helps the model capture temporal features from different perspectives, enabling effective exploration of change patterns in multitemporal RSI.

The MBCD process framework is shown in Fig. 6, where $\boldsymbol{F_i^1}$, $\boldsymbol{F_i^2}$, and $\boldsymbol{F_i^3} \in \mathbb{R}^{C\times H\times W}$ $(1 \le i \le 4)$ represent the image features extracted by the SSE in the same layer for three different time periods. Specifically, $Dim = C$ indicates concatenation along the channel (C) dimension for the three feature groups; $Dim = H$ denotes concatenation along the height (H) dimension; $Dim = W$ denotes concatenation along the width (W) dimension. By multidimensional concatenation of multitemporal features, three sets of richer deep semantic features can be generated. When concatenating in the width dimension, swapping the positions of $\boldsymbol{F_i^1}$ and $\boldsymbol{F_i^3}$ avoids redundancy in the features when concatenating in the height dimension, thereby obtaining a more diverse feature representation. As shown in the following equation:

$$\boldsymbol{F_i^C} = VSS\left(Concat\left(\boldsymbol{F_i^1}, \boldsymbol{F_i^2}, \boldsymbol{F_i^3}\right)_{dim=C}\right)$$
$$\boldsymbol{F_i^H} = VSS\left(Concat\left(\boldsymbol{F_i^1}, \boldsymbol{F_i^2}, \boldsymbol{F_i^3}\right)_{dim=H}\right)$$
$$\boldsymbol{F_i^W} = VSS\left(Concat\left(\boldsymbol{F_i^3}, \boldsymbol{F_i^2}, \boldsymbol{F_i^1}\right)_{dim=W}\right) \tag{7}$$
$$\boldsymbol{F_i} = Concat\left(\boldsymbol{F_i^C}, rearrange\left(\boldsymbol{F_i^H}\right), rearrange\left(\boldsymbol{F_i^W}\right)\right)_{dim=C}. \tag{8}$$

Through the above formulations, the three groups of concatenated features are processed by Conv and the VSS module, respectively, to obtain the features $\boldsymbol{F_i^C}$, $\boldsymbol{F_i^H}$, and $\boldsymbol{F_i^W}$. Since $1 \le i \le 4$, four sets of features can be obtained as shown in Fig. 6: $\boldsymbol{F_4}$, $\boldsymbol{F_3}$, $\boldsymbol{F_2}$, and $\boldsymbol{F_1}$.

Since the VSS Block uses 2-D-selective-scan for Vision Data, it can scan feature maps from multiple directions. Therefore, through three different feature concatenation methods and 2-D-selective-scan, multiphase features can form more directional scanning methods. In this way, VSS Block can effectively extract time information from features, thereby more accurately capturing the dynamic process of LCLU changes over time.

After that, the features are up-sampled using UP Sample, restoring the final output features to a higher resolution. Finally, the top-level feature $\boldsymbol{F_1}$ is generated into MBCM, thereby effectively reflecting changes in the surface at different points in time.

D. Multitemporal Learning Decoder

As shown in Fig. 7, this framework diagram illustrates the decoder architecture of the unified multitemporal learning framework. In the figure, $\boldsymbol{F_i^1}$, $\boldsymbol{F_i^2}$, and $\boldsymbol{F_i^3}(4 \ge i \ge 2)$ represent the multilevel features of the images at time T_1, time T_2, and time T_3, respectively. By inputting these three sets of

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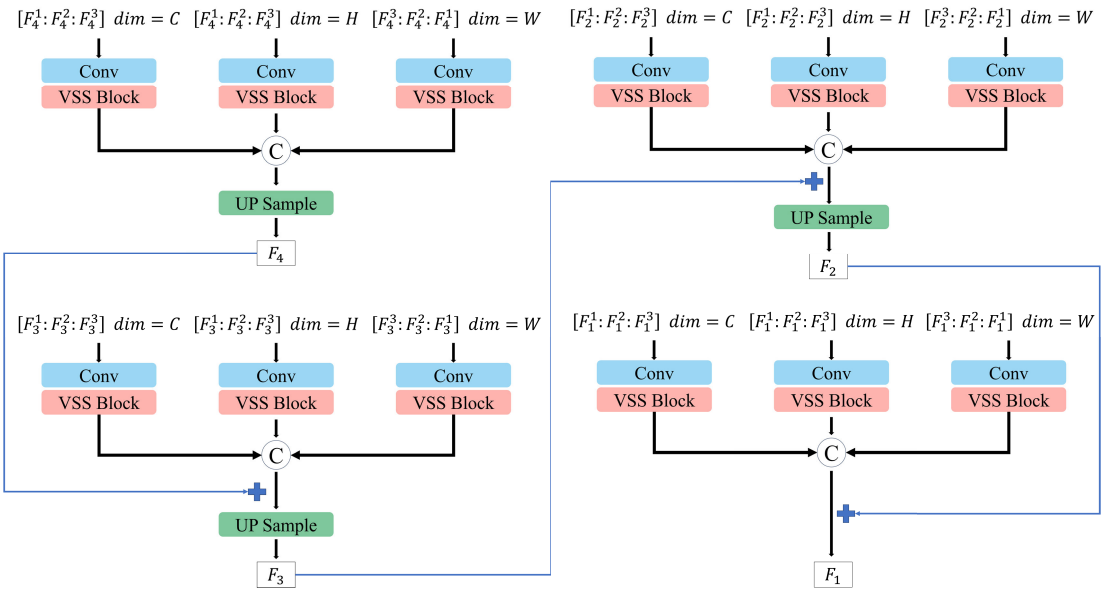


Fig. 6. Framework of MBCD block.

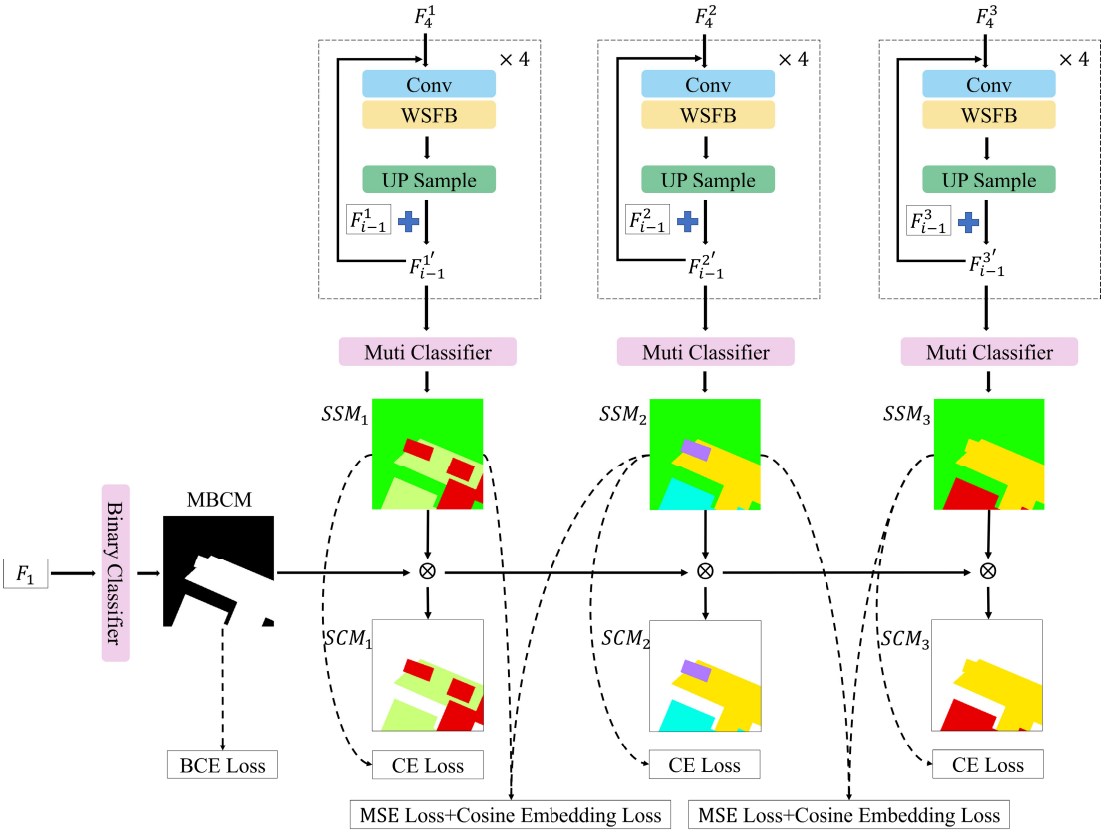


Fig. 7. MTLD of the unified multitemporal learning framework. The abbreviations WSFB and MBCM denote WSFB and multitemporal BCM.

features into the WSFB and UP Sample for iterative decoding. When i = 2, features $\boldsymbol{F_1^{1'}}$, $\boldsymbol{F_1^{2'}}$, and $\boldsymbol{F_1^{3'}} \in \mathbb{R}^{C\times H/4\times W/4}$ can be obtained. After processing them through a multiclassifier, three sets of SSMs X_1, X_2, and X_3 can be obtained. Generate MBCM by passing the fused feature $\boldsymbol{F_1}$ from MBCD through a binary classifier. Finally, X_1, X_2, X_3, and MBCM are multiplied together to mask the LCLU information of the unchanged areas and focus only on the information of the changed areas, thereby obtaining the SCMs Y_1, Y_2, and Y_3.

1) Wavelet Spatial-Frequency Block: In the decoder, WSFB is the core component of this module. Its structure is shown in Fig. 8, and it mainly consists of DWT, DW Conv, multihead self-attention (MSA), and VSS block.

WSFB can integrate spatial and frequency representations to achieve cross-domain fusion. First, dynamic position encoding is added to the input feature $\boldsymbol{F_{in}}$ to generate position encoding information. Next, VSS Block's time series modeling capabilities are used to generate spatial–temporal features.

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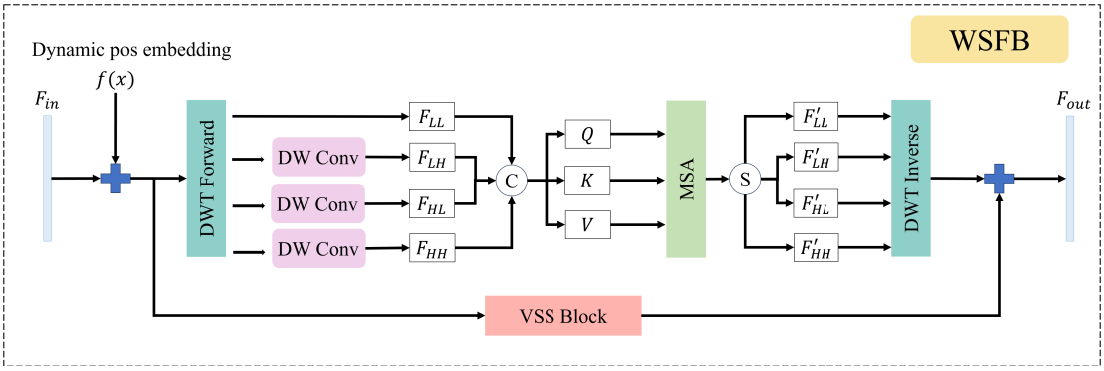


Fig. 8. WSFB.

Subsequently, the features are processed using DWT Forward to decompose them into multiple frequency bands ($\boldsymbol{F_{LL}}$, $\boldsymbol{F_{LH}}$, $\boldsymbol{F_{HL}}$, $\boldsymbol{F_{HH}}$), which effectively extracts low-frequency and high-frequency information at multiple scales. Pass the high-frequency band through DW Conv to further optimize the expression capability of features. Next, the frequency band features are concatenated, and long-range dependencies in the features are captured using MSA to further enhance the global information of the features. Then separate it into $\boldsymbol{F'_{LL}}$, $\boldsymbol{F'_{LH}}$, $\boldsymbol{F'_{HL}}$, and $\boldsymbol{F'_{HH}}$, and the processed sub bands are recombined using DWT Inverse. Finally, the reorganized features are added to the output features of the VSS Block to generate feature $\boldsymbol{F_{out}}$.

In WSFB, we use 2-D DWT to process features. As shown in (9), L and H are first generated by row direction filtering

$$\begin{aligned} \boldsymbol{L_r} &= DWT\left(\boldsymbol{f}, h_L\right) \\ \boldsymbol{H_r} &= DWT\left(\boldsymbol{f}, h_H\right) \end{aligned} \tag{9}$$

where f represents the input feature, and h_L and h_H represent low-pass/high-pass filters.

Then, through column direction filtering, F_{LL}, F_{LH}, F_{HL}, F_{HH} are generated

$$\begin{aligned} \boldsymbol{F_{LL}} &= DWT\left(\boldsymbol{L_r^T}, h_L\right)^T \\ \boldsymbol{F_{LH}} &= Conv\left(DWT\left(\boldsymbol{L_r^T}, h_H\right)^T\right) \\ \boldsymbol{F_{HL}} &= Conv\left(DWT\left(\boldsymbol{H_r^T}, h_L\right)^T\right) \\ \boldsymbol{F_{HH}} &= Conv\left(DWT\left(\boldsymbol{H_r^T}, h_H\right)^T\right). \end{aligned} \tag{10}$$

Finally, the features are recombined using the Inverse DWTs (IDWTs) according to

$$\begin{aligned} \boldsymbol{F_M} &= MSA\left(Concat\left(\boldsymbol{F_{LL}}, \boldsymbol{F_{LH}}, \boldsymbol{F_{HL}}, \boldsymbol{F_{HH}}\right)\right) \\ \boldsymbol{F'_{LL}}, \boldsymbol{F'_{LH}}, \boldsymbol{F'_{HL}}, \boldsymbol{F'_{HH}} &= Split\left(\boldsymbol{F_M}\right), \\ F_{out} &= IDWT_{row}\left(IDWT_{col}\left(\boldsymbol{F'_{LL}}, \boldsymbol{F'_{LH}}\right)\right. \\ &\qquad \left. IDWT_{col}\left(\boldsymbol{F'_{HL}}, \boldsymbol{F'_{HH}}\right)\right). \end{aligned} \tag{11}$$

DWT can decompose input signals (or features) into different frequency bands, namely low-frequency and high-frequency components. The low-frequency component mainly contains the overall information of the image (such as smooth areas and global features). The high-frequency component reveals the detailed information in the image (such as edges and noise). The characteristics of this filtering make DWT effective in capturing both global and local features of images.

Since DWT provides feature inputs from different frequency bands. MSA can not only capture long-range dependencies of features, but also correlate contextual relationships at different scales through information from various frequency bands.

2) Joint Multiloss Supervision (Loss Function): During model training, a joint loss function is used to guide the generation of Y_1, Y_2, Y_3, and MBCM in Fig. 7. The loss function mainly consists of cross-entropy loss L_{scd}, binary cross-entropy loss L_{bcd}, mean squared error loss L_{mse}, and cosine distance loss L_{cos}. Here, we take three temporal phases as examples, and the definition of loss is as follows:

$$Loss = \underbrace{L_{scd} + L_{bcd}}_{ce\ \ loss} + \underbrace{\left(L_{mse} + L_{cos}\right)}_{margin\ \ loss}. \tag{12}$$

L_{scd} can be used to generate the SSMs (X_1, X_2, and X_3) in Fig. 8, which are defined as follows:

$$L_{scd} = -\frac{1}{n}\sum_{i=1}^{n} y_i log\hat{y}_i \tag{13}$$

where n represents the number of classes, y_i and $\hat{y}_i$ represent the true labels and predicted probability values, respectively.

L_{bcd} can be used to generate MBCM, which is calculated as follows:

$$L_{bcd} = -\left[y log\left(\hat{y}\right) + (1-y)\, log\left(1-\hat{y}\right)\right] \tag{14}$$

where y and $\hat{y}$ represent the true label and predicted probability value, respectively (when $y = 0$ indicates no change, and $y = 1$ indicates change).

L_{mse} is defined as follows:

$$\begin{aligned} L_{mse} &= \frac{1}{2}\left(L_{mse}^{X_1 \to X_2} + L_{mse}^{X_2 \to X_3}\right) \\ L_{mse}^{X_1 \to X_2} &= \frac{1}{N}\sum_{i=1}^{n}\left(X_{1i} - X_{2i}\right)^2 * y_{X_1 \to X_2} \\ L_{mse}^{X_2 \to X_3} &= \frac{1}{N}\sum_{i=1}^{n}\left(X_{2i} - X_{3i}\right)^2 * y_{X_2 \to X_3}. \end{aligned} \tag{15}$$

The label $y_{X_1 \to X_2}$ indicates that the semantic label values of X_1 and X_2 are XNOR. The label $y_{X_2 \to X_3}$ indicates that the semantic label values of X_2 and X_3 are XNOR (the *change* area is 0, and the *unchanged* area is 1). Therefore, L_3 masks semantically different areas in the two feature sets and reduces the feature distance between areas with the same semantics.

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TABLE I
ABLATION EXPERIMENT RESULTS ON THE HI-UCD AND WUSU DATASETS

Dataset	Method	MBCD Block	WSFB	OA(%)	mIoU(%)	Sek(%)	F_{scd}(%)
Hi-UCD	Base Mamba			91.60	79.72	28.05	58.35
	W/MBCD	✓		**91.62**	80.54	29.38	58.99
	W/WSFB		✓	91.60	80.81	29.42	58.81
	Ours	✓	✓	91.49	**81.08**	**29.73**	**59.06**
WUSU	Base Mamba			88.84	75.75	14.52	45.75
	W/MBCD	✓		89.18	75.87	14.68	45.75
	W/WSFB		✓	88.99	75.33	14.63	46.35
	Ours	✓	✓	**89.33**	**75.99**	**15.07**	**46.42**

L_{cos} is used to enhance the semantic consistency between X_1, X_2, and X_3, and its formula is as follows:

$$L_{cos} = \frac{1}{2}\left(L_{cos}^{X_1 \to X_2} + L_{cos}^{X_2 \to X_3}\right)$$
$$L_{cos}^{X_1 \to X_2} = \begin{cases} 1 - cos\left(X_1, X_2\right), & y_{X_1 \to X_2} = 1 \\ cos\left(X_1, X_2\right), & y_{X_1 \to X_2} = 0 \end{cases}$$
$$L_{cos}^{X_2 \to X_3} = \begin{cases} 1 - cos\left(X_2, X_3\right), & y_{X_2 \to X_3} = 1 \\ cos\left(X_2, X_3\right), & y_{X_2 \to X_3} = 0. \end{cases} \quad (16)$$

Overall, the proposed loss function provides complementary supervision for semantic representation learning from different levels. L_{scd} supervises the SS outputs of each single-temporal image with one-hot semantic labels, guiding the model to learn discriminative land-cover representations for individual temporal phases. L_{bcd} supervises the BCM using the predicted probabilities of changed and unchanged regions, encouraging the MBCD module to capture change-localization information from multitemporal features. In addition, L_{mse} and L_{cos} impose temporal feature consistency constraints, reducing the feature distance between regions with the same semantic category across temporal phases while enlarging the distance between features belonging to different semantic categories. Therefore, these loss terms jointly constrain single-temporal semantic discrimination, multitemporal change localization, and temporal feature consistency, which is consistent with the semantic evolution pattern of multitemporal SCD.

IV. EXPERIMENT

A. Datasets

1) Hi-UCD Mini Dataset [26]: The Hi-UCD mini dataset was captured in parts of Tallinn, the capital of Estonia, covering an area of approximately 30 km^2. The images were captured by ADS100-SH100 cameras in 2017, 2018, and 2019, respectively. The spatial resolution of the images in this dataset is 0.1 m, and it contains a total of nine LCLU categories. The image resolution is 1024×1024 pixels, with a total of 359 pairs of images between 2017 and 2018, and 386 pairs of images between 2018 and 2019.

During the data processing stage, we first extracted data samples with images from the same region covering three time periods, obtaining a total of 204×3 images. Subsequently, using the corresponding semantic label, generate a binary change label and a semantic change label. However, since binary change labels are generated from semantic labels, discontinuities may occur at LCLU boundaries. In addition, LCLU position shifts caused by differences in shooting times may result in misalignments of label boundaries. Therefore, we compared the binary change labels with the original images and performed manual secondary annotation to improve the annotation accuracy.

Since the size of a single image is 1024×1024, it is not suitable for model training. We divide it into small blocks of 256×256. A total of 3264 images were obtained. Considering the small amount of data, we divided the dataset into a training set and a test set in an 8:2 ratio.

2) WUSU Dataset [27]: The images in the WUSU dataset were mainly collected in key development areas such as Jiang'an District and Hongshan District in Wuhan, China, covering an area of approximately 80 km^2. The images were acquired by the GF-2 satellite in 2015, 2016, and 2018, respectively, with a maximum cloud cover of 5%. This dataset consists of 4-band fusion satellite imagery with a spatial resolution of 1 m, containing a total of 11 LCLU categories.

Similarly, during the data processing stage, we organized the image data. Considering that there were many duplicate images in the dataset, we deduplicated them during the organization process. Subsequently, we addressed the binary change labels generated by the semantic labels. We combined the original images with manual secondary annotations to correct annotation errors caused by boundary discontinuities or positional shifts.

The single image size of the data is 512×512, but approximately half of the pixel area is repeated during shooting. Therefore, we divide the images into 256×256 blocks and remove duplicate areas. Finally, we divide the dataset into training, validation, and test sets in a 5:2:3 ratio.

3) DynamicEarthNet Dataset [15]: This dataset contains multitemporal remote sensing imagery from 75 regions worldwide, covering the period from 2018 to 2019 (captured daily). Each area contains a total of 750 images with a spatial resolution of 3 m. It also provides semantic annotation data for the first day of each month, with a total of 24 semantic coverage maps for each region.

To better suit the objectives of this study, we followed [45] and selected imagery from the first day of every fourth month, thereby constructing a six-time-phase dataset for each region to meet the experimental requirements for multitemporal SCD. The original image size is 1024×1024. Each image was cropped into four 512×512 patches. The time-series samples from 75 regions were then divided into training, validation, and test sets in a ratio of 7:2:1.

B. Experimental Setup

The framework is implemented based on PyTorch and is trained and tested on a hardware platform equipped with two NVIDIA GeForce RTX 3080 Ti GPUs. A total of 50 epochs

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were set during training, with a batch size of 2 (i.e., 2 × 3 images were processed simultaneously in each batch). The initial learning rate is 1e-4, and the learning rate decay rate is 5e-4 to ensure optimization during training. The optimizer selected is Adam to accelerate convergence and improve model stability.

C. Evaluation Criteria

This article uses four metrics to evaluate the performance of multitemporal SCD, namely overall accuracy (OA), mean intersection over union (mIoU), separation kappa (SeK) coefficient, and F_{SCD}. mIoU mainly evaluates the results of MBCD, while SeK and F_{SCD} mainly evaluate the results of SCD.

OA is defined as follows:

$$OA = \sum_{i=0}^{N} q_{ii} \Big/ \sum_{i=0}^{N}\sum_{j=0}^{N} q_{ij} \tag{17}$$

where $Q = \{q_{ij}\}$ is the confusion matrix, N is the number of classes.

The definition of mIoU is as follows:

$$\begin{aligned} mIoU &= (IoU_{nc} + IoU_C) \Big/ 2 \\ IoU_{nc} &= q_{00} \Big/ \left(\sum_{i=0}^{N} q_{i0} + \sum_{i=0}^{N} q_{0j} - q_{00} \right) \\ IoU_c &= \sum_{i=1}^{N}\sum_{j=1}^{N} q_{ij} \Big/ \left(\sum_{i=0}^{N}\sum_{j=0}^{N} q_{ij} - q_{00} \right). \end{aligned} \tag{18}$$

SeK primarily evaluates the performance of models in detecting semantic change categories. It is defined as follows:

$$\begin{aligned} \alpha &= \sum_{i=0}^{N} \hat{q}_{ii} \Big/ \sum_{i=0}^{N}\sum_{j=0}^{N} \hat{q}_{ij} \\ \beta &= \sum_{i=0}^{N} \left(\sum_{j=0}^{N} \hat{q}_{ij} * \sum_{j=0}^{N} \hat{q}_{ji} \right) \Big/ \left(\sum_{i=0}^{N}\sum_{j=0}^{N} \hat{q}_{ij} \right)^2 \\ SeK &= e^{IoU_c - 1} \cdot (\alpha - \beta) \Big/ (1 - \beta) \end{aligned} \tag{19}$$

where $\hat{Q} = \{\hat{q_{ij}}\}$ is the confusion matrix and $\hat{q}_{ij} = q_{ij}$.

F_{scd} effectively captures the semantic class recognition ability of the model. It is defined as follows:

$$\begin{aligned} P_{scd} &= \sum_{i=1}^{N} q_{ii} \Big/ \sum_{i=1}^{N}\sum_{j=0}^{N} q_{ij} \\ R_{scd} &= \sum_{i=1}^{N} q_{ii} \Big/ \sum_{i=0}^{N}\sum_{j=1}^{N} q_{ij} \\ F_{scd} &= \frac{2 * P_{scd} * R_{scd}}{P_{scd} + R_{scd}}. \end{aligned} \tag{20}$$

D. Ablation Studies

1) Effectiveness Experiment of MBCD Block: To validate the effectiveness of multidimensional feature concatenation in the MBCD block, ablation experiments of Base Mamba and W/MBCD were designed as shown in Table I. Base Mamba indicates that only Mamba is used as the backbone network in the entire framework, without using the MBCD block and WSFB. W/MBCD indicates that the MBCD block has been added to the Base Mamba. By comparing the results of these two experiments, the effectiveness of the MBCD block can be verified.

As shown in Table I, W/MBCD significantly outperforms Base Mamba in all evaluation metrics on two multitemporal datasets. This result indicates that the MBCD block enriches the temporal relationship of features by concatenating multitemporal features. Enabling the model to better capture time-dependent relationships in multitemporal images, thereby enhancing the model's ability to perceive changes in the temporal dimension of RSI. Therefore, compared with Base Mamba, W/MBCD shows a significant performance improvement, fully proving the effectiveness of the MBCD block.

2) Effectiveness Experiment of WSFB: To validate the effectiveness of WSFB, ablation experiments were designed for Base Mamba and W/WSFB, as shown in Table I. W/WSFB refers to Base Mamba using only WSFB. By comparing the results of these two experiments, the effectiveness of this module can be effectively validated.

As shown in Table I, W/WSFB significantly outperforms Base Mamba in all evaluation metrics on these two datasets. Since WSFB effectively combines DWT and MSA, the model can decompose features into low-frequency and high-frequency components. This ensures that MSA can capture the contextual relationships between features at different frequencies through information across various frequency bands. Therefore, W/WSFB has improved in all performance metrics, proving the effectiveness of WSFB in enhancing the model's ability to capture spatial–temporal dependencies and improving model performance.

To further analyze the contribution of the wavelet transform in WSFB, we replaced the WSFB with a spatial-channel attention (SCA) block of similar parameters.

As shown in Table II, the performance decreases when WSFB is replaced by SCA, indicating that the performance gain is not solely from the attention mechanism but also from the wavelet-based frequency decomposition. The wavelet transform enables the model to capture both low-frequency structural information and high-frequency details, which is beneficial for fine-grained CD.

3) Effectiveness Experiment of VSS: To further evaluate the effectiveness of the proposed VSS module, we replaced all VSS modules in Base Mamba with the self-attention mechanism of the Temporal Transformer, naming the resulting model the base transformer.

The results in Table III show that the Mamba-based VSS module achieves consistently better performance across multiple metrics. This improvement can be attributed to its linear computational complexity and stronger ability to model long-range temporal dependencies compared to the $O(N^2)$ complexity of self-attention. Moreover, while Transformer relies on global attention and may suffer from overfitting

TABLE II
COMPARISON RESULTS OF WSF BLOCK AND SCA BLOCK

Dataset	Method	OA (%)	mIoU (%)	Sek (%)	F_{scd} (%)
Hi-UCD	SCA Block	90.19	76.03	26.09	60.51
	Ours (WSF Block)	91.49	81.08	29.73	59.06
WUSU	SCA Block	86.66	66.64	11.90	43.63
	Ours (WSF Block)	89.33	75.99	15.07	46.42

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TABLE III
COMPARISON RESULTS OF MAMBA AND TRANSFORMER BASE MODELS

Dataset	Method	OA (%)	mIoU (%)	Sek (%)	F_{scd} (%)
Hi-UCD	Base Transformer	91.28	74.62	21.18	55.88
	Base Mamba	91.60	79.72	28.05	58.35
WUSU	Base Transformer	87.51	66.93	8.36	40.18
	Base Mamba	88.84	75.75	14.52	45.75

TABLE IV
EXPERIMENTAL RESULTS ON THE EFFECTIVENESS OF METRIC LOSS ON THE HI-UCD DATASET

Method	L_{mse}	L_{cos}	OA(%)	mIoU(%)	Sek(%)	F_{scd}(%)
Wo L_{mse}/L_{cos}			90.90	79.76	27.75	57.37
Wo L_{mse}		✓	90.27	78.76	25.76	56.53
Wo L_{cos}	✓		90.59	78.15	25.60	56.81
Base Mamba	✓	✓	**91.60**	**79.72**	**28.05**	**58.35**

TABLE V
EXPERIMENTAL RESULTS ON THE EFFECTIVENESS OF METRIC LOSS ON THE WUSU DATASET

Method	L_{mse}	L_{cos}	OA(%)	mIoU(%)	Sek(%)	F_{scd}(%)
Wo L_{mse}/L_{cos}			88.64	75.28	14.37	45.42
Wo L_{mse}		✓	88.65	75.07	14.08	45.45
Wo L_{cos}	✓		88.52	74.74	13.63	45.05
Base Mamba	✓	✓	**88.84**	**75.75**	**14.52**	**45.75**

under limited training data, Mamba adopts a selective scanning mechanism to dynamically model temporal information, enabling more flexible adaptation to varying temporal patterns and better generalization performance.

4) Effectiveness Experiment of Metric Loss: Since the cross-entropy loss function is indispensable in SS tasks, this section mainly verifies the effectiveness of the metric losses L_3 and L_4. As shown in Tables II and III, Wo L_3/L_4 indicates L_3 and L_4 is not used in Base Mamba; Wo L_3 indicates that $L4$ is not used in Base Mamba; Wo L_4 indicates that L_4 is not used in Base Mamba; Base Mamba indicates optimization using L_3 and L_4. The results of this ablation experiment can be used to verify the effectiveness of the metric loss.

As shown in Tables II and III, Base Mamba achieves the best results in all evaluation metrics. This is because the method constrains the distance between samples in the feature space from different perspectives through two different metric losses. This mechanism enhances the model's understanding of the internal structure of categories, ensuring that the model can accurately distinguish between categories even when there are few samples. Therefore, Base Mamba has improved in all performance metrics, fully demonstrating the important role of metric losses L_3 and L_4 in improving model performance in few-shot learning.

E. Comparative Experiments

1) Comparative Methods: To validate the effectiveness of the framework proposed in this article, we selected three dual-phase SCD frameworks as comparison experiments. As shown in Fig. 2, these are HRSCD-str.4 [11], SSCD-l [28], and TED [29], respectively. Since all three sets of frameworks use bi-temporal RSI, the first and third temporal images were selected for training and testing in this experiment. The effectiveness of the multitemporal SCD framework was verified by comparing it with the bi-temporal SCD framework method.

Finally, we improved upon ChangeMamba [40] and constructed Base Mamba, a comparison method suitable for multiphase SCD. Through comparative experiments with Base Mamba, the performance advantages of the proposed model can be effectively verified.

2) Experiments on the Hi-UCD Dataset: The experimental results of the Hi-UCD dataset are shown in Fig. 9. First, in the three temporal RSI of Fig. 9(a1)–(a3), the main changes occurred between (a1) and (a2), while (a2) and (a3) remained largely unchanged. The results based on the dual-phase framework performed poorly overall: HRSCD-str.4 failed to detect some grassland changes and falsely detected building changes below Fig. 9(a1); SSCD-l and TED both produced false detections near the change areas. In the multiphase detection framework, Base Mamba has missed detections and failed to capture continuous grassland changes. The method proposed in this article can identify the changed areas more accurately. Since there is no change from Fig. 9(a2) to (a3), it can be regarded as a single bi-temporal SCD. This provides a good scenario for verifying the performance of this method in bi-temporal SCD tasks. The experimental results also prove that the method proposed in this article is superior to traditional bi-temporal frameworks.

Changes in the intermediate transition state appeared in images Fig. 9(b1)–(b3). The results based on the bi-temporal framework detected large areas of unchanged regions due to the lack of Fig. 9(b2) temporal images. The Base Mamba model produces false detections for changes between roads and bare land. Although the proposed method exhibits a small number of false detections along change boundaries, it achieves higher overall detection accuracy.

In Fig. 9(c1)–(c3), the main change is the transition from water to no water to water inside the pool. The results based on the bi-temporal framework failed to identify changes. Base Mamba had a large area of false positives around the change area, while our method performed best in change localization and shape preservation.

The quantitative experimental results on the Hi-UCD dataset are shown in Table VI. The experimental performance of the SCD framework based on bi-temporal RSI was poor, with all metrics falling below the experimental results based on the multitemporal framework. Although Base Mamba's results are relatively high, they are still lower than our method. Our method achieved good experimental results in the key metrics of mIoU, Sek, and F_scd, reaching 81.08%, 29.73%, and 59.06%, respectively. This result indicates that our method can more accurately capture change information, thereby demonstrating stronger performance and adaptability in SCD tasks.

3) Experiments on the WUSU Dataset: The qualitative experimental results of the WUSU dataset are shown in Fig. 10. First, in the set of multitemporal RSI of Fig. 10(a1)–(a3), there is no obvious change between (a1) and (a2). However, there is a significant change between Fig. 10(a2) and (a3), mainly involving a transition from buildings to construction sites. All three methods based on the bi-temporal framework showed large areas of missed detection and false detection. In the multitemporal detection framework, Base Mamba performs better than the bi-temporal method. It can detect both high-rise and low-rise buildings in Fig. 10(a1) and (a2). However, the detection performance in Fig. 10(a3)

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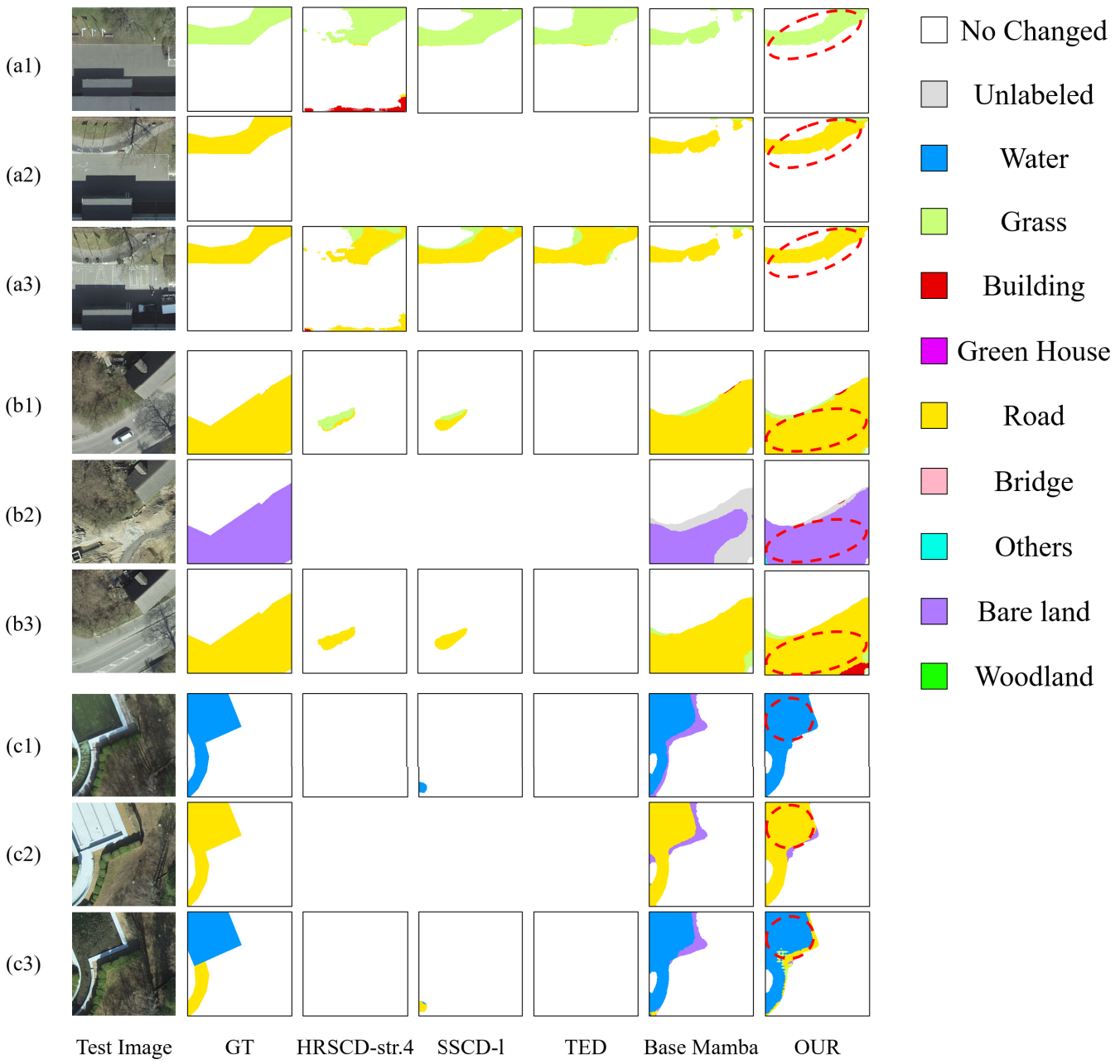


Fig. 9. Experiment result on Hi-UCD dataset. (a1)–(a3), (b1)–(b3), and (c1)–(c3) Three sets of multitemporal images, each corresponding to the before, mid, and after stages of the semantic LCLU changes.

is still inferior to our method, with some construction sites being falsely detected as grasslands.

In Fig. 10(b1)–(b3), continuous changes occurred. The detection performance of the bi-temporal framework was generally unsatisfactory, while the multitemporal detection framework was more accurate in locating changes and identifying change types. Although there were still a few false positives, the main changes were detected correctly.

In Fig. 10(c1)–(c3), HRSCD-str.4, SSCD-l, and TED were able to identify areas of change, but there were many false positives in the classification of change types. For example, in Fig. 10(c3), large areas of bare land were misclassified as buildings. Base Mamba outperforms the above three methods in Fig. 10(c3) results. However, its detection performance for Fig. 10(c1) is still inferior to ours, as it falsely detects some grassland areas as construction sites. Our method achieves higher accuracy in identifying complex changes by performing temporal modeling on multitemporal RSI and introducing WSFB to enhance the ability to extract temporal features.

The quantitative experimental results on the WUSU dataset are shown in Table VI. The experimental results of the SCD method based on the bi-temporal framework are lower than those based on the multitemporal framework. Our method achieved the best results in terms of OA, mIoU, Sek, and F_scd. In particular, there were significant improvements in the Sek and F_scd metrics compared to the comparison methods, reaching 15.07% and 46.42%, respectively. This further validates the effectiveness of our method. By combining WSFB to interactively model low-frequency and high-frequency information

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TABLE VI
COMPARISON RESULTS ON THE HI-UCD AND WUSU DATASETS

Dataset	Temporal	Method	OA (%)	mIoU (%)	Sek (%)	F_{scd} (%)
Hi-UCD	Bi-temporal	HRSCD-str.4	89.43	72.29	16.07	49.50
		SSCD-l	**92.22**	78.00	25.48	58.44
		TED	91.86	77.36	25.08	58.46
	Multi-temporal	Base Mamba	91.60	79.72	28.05	58.35
		Ours	91.49	**81.08**	**29.73**	**59.06**
WUSU	Bi-temporal	HRSCD-str.4	71.20	53.73	0.24	13.11
		SSCD-l	83.61	65.96	4.99	27.81
		TED	81.52	63.43	5.24	28.00
	Multi-temporal	Base Mamba	88.84	75.75	14.52	45.75
		Ours	**89.33**	**75.99**	**15.07**	**46.42**

TABLE VII
RESULT OF "FROM TO" ON THE HI-UCD AND WUSU DATASETS. RED DASHED LINES ARE USED IN THE HI-UCD DATASET, AND BLUE DASHED LINES ARE USED IN THE WUSU DATASET

Dataset	Temporal	LCLU Change
Hi-UCD	(a1)→(a2)→(a3)	Grass → Road → Road
	(b1)→(b2)→(b3)	Road → Bare land → Road
	(c1)→(c2)→(c3)	Water → Road → Water
WUSU	(a1)→(a2)→(a3)	High building → High building → Excavation
	(b1)→(b2)→(b3)	Woodland/Low building → Excavation → High building
	(c1)→(c2)→(c3)	Grassland → Excavation → High building/Bare surface

of image features, we can capture change information more accurately. This enables our model to capture subtle changes more sensitively when processing multitemporal data, further improving the performance of SCD.

In summary, because the Hi-UCD dataset has higher resolution and a relatively small number of classes, the performance gap between different methods is narrowed. In the WUSU dataset, which has lower resolution and more complex classes, the advantages of our method in feature modeling and handling class imbalance are even more pronounced. This results in a more significant performance improvement.

F. From-To Semantic Change Analysis

Table VII summarizes the "from-to" semantic change trajectories across different temporal phases on the Hi-UCD and WUSU datasets. Specifically, each sequence describes the evolution of land-cover categories from one time phase to the next, thereby explicitly capturing the dynamic change process rather than only the final result.

For example, as shown by the red dashed regions in Fig. 9 for the Hi-UCD dataset, transitions such as "Grass" → "Road" → "Road" and "Road" → "Bare land" → "Road" reveal evolutionary patterns from stability to transition and then recovery. In the WUSU dataset, as illustrated by the blue dashed regions in Fig. 10, sequences such as "Woodland/Low building" → "Excavation" → "High building" demonstrate more complex and progressive land-use transformations.

These results indicate that the proposed method is capable of effectively modeling fine-grained semantic evolution under multitemporal settings, thereby overcoming the limitations of traditional bi-temporal "change/no-change" representations and enabling a more comprehensive characterization of temporal change patterns.

G. Generalization Validation Experiment

As shown in Fig. 11, in the two six temporal phases, the detection results using the bi-temporal change method are poor, often failing to identify many areas of change. In multitemporal methods, both the Base Mamba and our method achieve good detection results. In contrast, our method is more sensitive to areas of change and can capture change information more comprehensively, thereby achieving better visualization results.

As shown in Table VIII, our method still achieved the best results across multiple evaluation metrics. This indicates that the method is not only suitable for tri-temporal scenarios but can also be effectively extended to modeling tasks involving longer time series, demonstrating strong generalization capabilities. Although the mIoU score is relatively low, this is primarily due to the more complex class distribution in the six temporal phases scenario. Furthermore, the bi-temporal method tends to classify most regions as unchanged, which has somewhat affected the performance of this metric.

In summary, experiments on the six temporal phases dataset further demonstrate the effectiveness and advantages of the proposed method in long-term temporal scenarios.

H. Computational Cost Comparison Experiment

We measure model complexity by the number of parameters (Params) and floating-point operations per second (FLOPS). Input uniformly uses bi-temporal, tri-temporal and six-time-phase 512 × 512 pixel images.

As shown in Fig. 12, the multitemporal method is higher than the bi-temporal method in both Params and FLOPS metrics. This is because multitemporal methods require processing more image features, which in turn increases the amount of computation required. In multitemporal methods, our method has a higher computational cost than Base Mamba. This is due to the additional introduction of the MBCD and WSF Block.

When our model was expanded from three-time phases to six, the number of model parameters did not double despite the doubling of the number of time phases; instead, it increased only slightly. The increase in FLOPs is relatively

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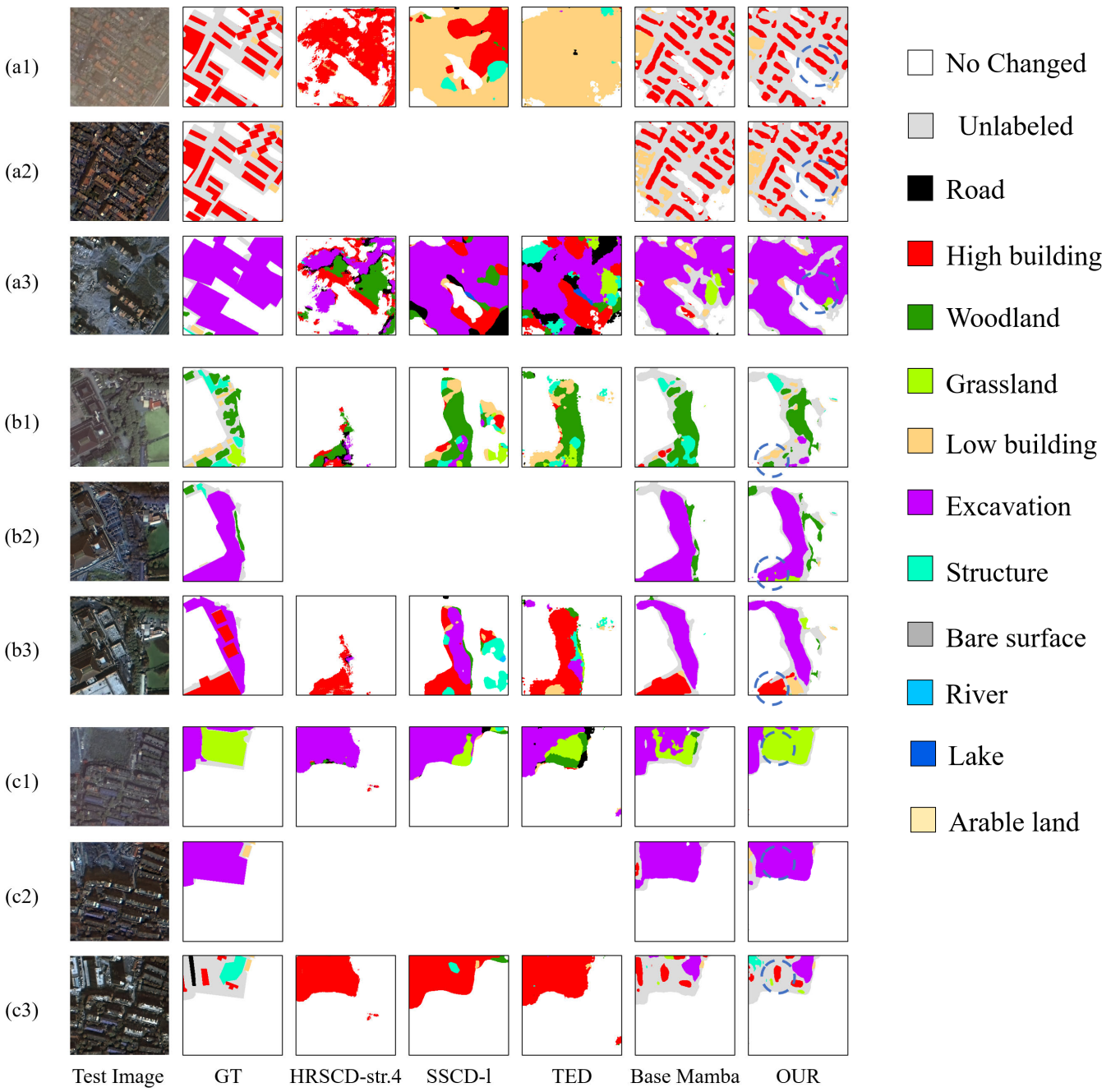


Fig. 10. Experiment result on WUSU dataset. (a1)–(a3), (b1)–(b3), and (c1)–(c3) Three sets of multitemporal images, each corresponding to the before, mid, and after stages of the semantic LCLU changes.

TABLE VIII
COMPARISON RESULTS ON THE DYNAMICEARTHNET DATASET

Temporal	Method	OA (%)	mIoU (%)	Sek (%)	F_{scd} (%)
Bi-temporal	HRSCD-4	58.37	47.08	24.07	60.13
	SSCD-I	58.79	53.08	25.14	60.18
	TED	59.26	**55.71**	25.07	60.53
Multi-temporal	Base Mamba	67.67	54.83	46.14	70.13
	Ours	**72.19**	52.42	**53.24**	**75.15**

more pronounced, as the forward pass requires processing feature maps from more temporal phases, thereby increasing the computational load. Consequently, the model demonstrates good scalability in terms of parameter size.

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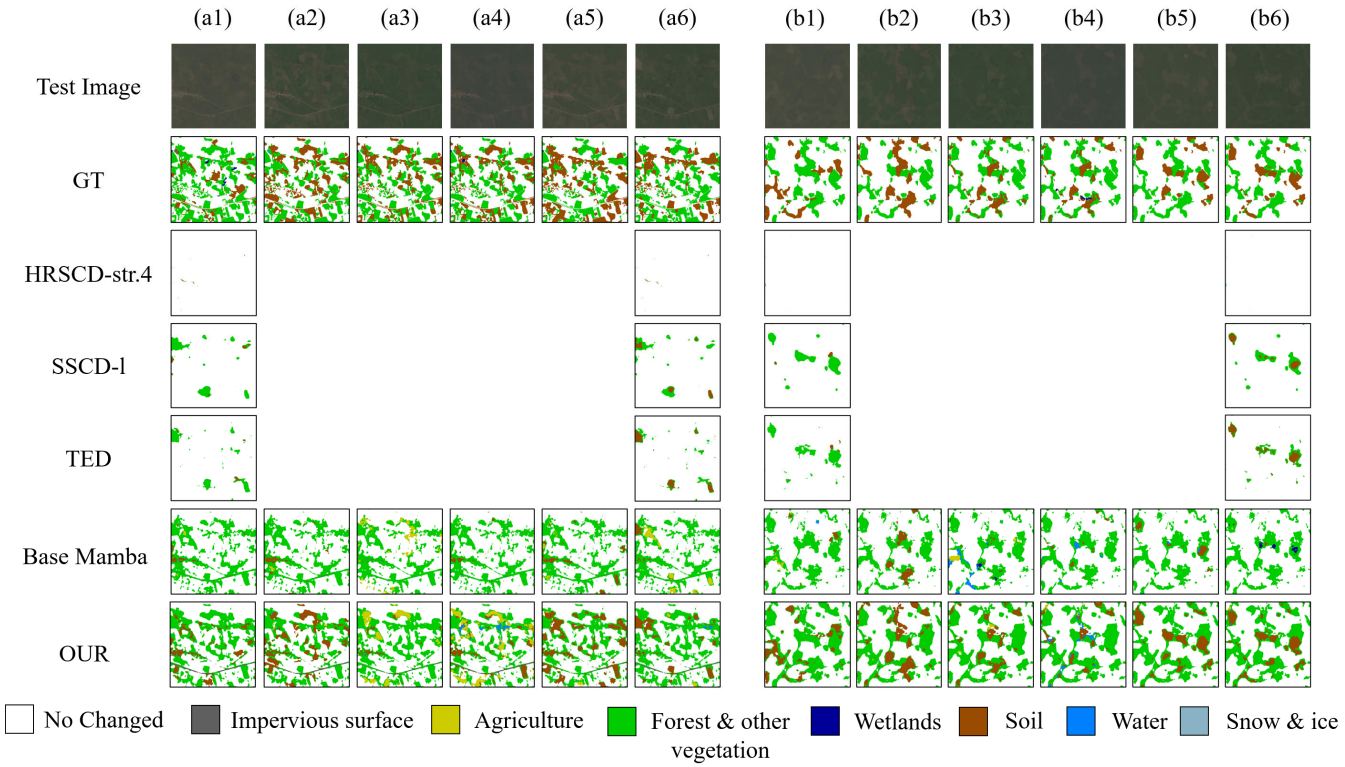


Fig. 11. Experiment result on the DynamicEarthNet dataset.

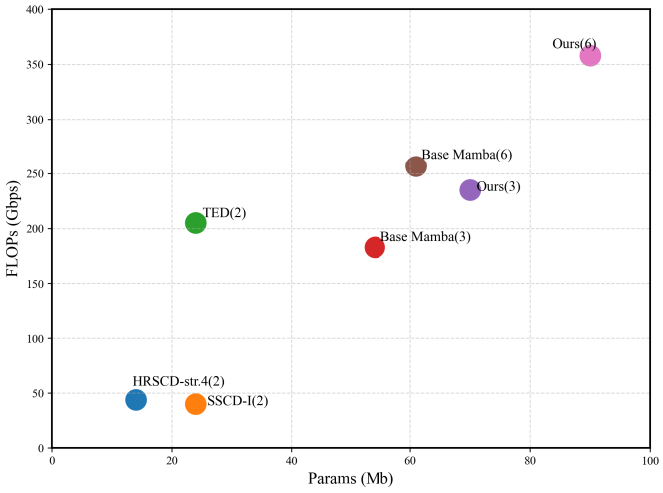


Fig. 12. Model efficiency comparison results. The horizontal axis represents the number of Parameters (Params.) and the vertical axis represents the number of FLOPS.

Furthermore, although the computational load increased on the six-phase dataset, the model still achieved the best results. This indicates that the proposed framework maintains acceptable computational overhead while delivering strong performance.

V. Conclusion

This article proposes a unified multitemporal learning framework. It splits the SCD task into two subtasks, SS and MBCD, and combines multiple loss functions for collaborative optimization to effectively improve the model's learning ability. The framework consists mainly of three core modules: encoder, decoder, and MBCD block. The encoder maps multitemporal RSI to a high-dimensional spatial–temporal feature space. WSFB is introduced in the decoder to capture cross-band long-range dependencies and map high-dimensional features to SSMs. The MBCD module generates MBCM by integrating multidimensional multitemporal features. Finally, by integrating multitemporal SSMs with MBCM, multitemporal SCMs can be obtained.

In the future, we will further explore SCD in multitemporal RSI. This includes extending to longer time scales to reveal long-term evolution patterns and integrating multisource remote sensing data of different resolutions and bands to enhance data diversity.

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